

- 1. Define Artificial Intelligence (AI)**
- 2. Discuss the impact of AI on different sectors such as healthcare, finance, and transportation.**
- 3. Identify and explain at least five key characteristics of Artificial Intelligence.**
- 4. Choose a specific AI application and analyze its working principles, benefits, and potential ethical concerns.**
- 5. Define an Intelligent Agent and explain its role in Artificial Intelligence.**
- 6. Discuss the key properties that characterize an intelligent agent.**
- 7. Describe the functions of sensors, actuators, and effectors in an intelligent agent.**
- 8. Explain the concept of an environment in which an intelligent agent operates.**
- 9. Explain the concept of a rational agent. Explain its types.**
- 10. Explain the concept of state space search in problem-solving.**
- 11. Describe the process of problem formulation in artificial intelligence.**
- 12. Compare and contrast well-defined problems and constraint satisfaction problems (CSPs).**
- 13. Analyze the importance of heuristics in AI problem-solving.**
- 14. Compare and contrast Depth-First Search (DFS) and Breadth-First Search (BFS).**
- 15. Explain the concept of Depth-Limited Search (DLS).**
- 16. Conduct a detailed comparison of search strategies: DFS, BFS, and DLS.**
- 17. Describe the Best-First Search strategy and how it differs from uninformed search techniques.**
- 18. Discuss the A Search algorithm in detail.***
Explain the heuristic function and how it influences the search process. Provide an example to illustrate the working of A Search.*
- 19. Explain the Minimax procedure in adversarial search.**
- 20. Describe the Alpha-Beta Pruning technique and how it improves the efficiency of the Minimax algorithm.**
- 21. Explain expert system with its structure.**